

Design and User Experience Evaluation of 3D Product Information in XR Shopping Application

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ABSTRACT

Current 3D product information pages are believed to enrich the online shopping experience since it provides a more immersive experience. However, existing solutions still retain the 2D UI elements in the product information presentation design, preventing users from fully immersing themselves in the virtual environment and further degrading the shopping experience. In order to evaluate the user experience of 3D product information in XR shopping applications, we first construct a design space based on previous design cases of product information presentation in virtual environments and produce nine new solutions by combining elements in the design space.

Index Terms: Human-centered computing—Visualization—Visualization design and evaluation methods—; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Extended Reality (XR) technologies are considered very promising technological tools, able to produce satisfactory consumer experiences [4]. It has previously been observed that because the depth information is increased in the XR online shopping environment, the information presentation of products can get rid of the limitation of a 2D interface and with a more diverse and immersive expression. [5]. Although numerous e-commerce platforms have designed various solutions of displaying product information in XR applications, these solutions still have many deficiencies. On the one hand, some applications use 2D graphics which superimpose in a three-dimensional environment or 2D static pop-ups to present the products' information. For a 3D interactive shopping environment, 2D information can cause a fragmented flow experience for users when evaluating or viewing products; on the other hand, some applications utilize a range of different information design when showing product information, resulting in illogical and disorderly page layouts that lead to a longer learning curve and cognitive load for users. These cases support the view that current 3D information presentation solutions have several outstanding issues that significantly impact the user's shopping experience.

In our work, we first deconstruct and analyze several existing 3D product information presentation design cases. Then construct a design space and apply elements in the design space for different combinations to produce some new solutions. The aim of this work is to explore the different impacts of different dimensions and different combination elements in the design space on the user's shopping experience, and to provide suggestions and references for the design of future 3D product information.

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2 DESIGN SPACE

We construct a design space using a collection of design cases based on an organizational taxonomy in this chapter.

2.1 Selecting Design Cases

In order to develop a collection of design cases, we consider extensively searching current 3D product information presentation solutions in XR shopping applications in search engines like Google and Bing. Additionally, to ensure that the design cases of 3D information presentation solutions collected are representative, we need to establish the inclusion criteria. Specifically, the information presentation solutions in these design cases need to have three characteristics: interactivity, information and immersion. We obtained 50 different design cases representing a selection of designs we had made specifically for presenting 3D information in XR shopping applications.

2.2 Deconstructing Design Cases And Constructing The Design Space

Our research use an open coding approach referring to the work of Ens et al. [2] to analyze and deconstruct the design cases in the collection according to the factors described in the following paragraph.

We observed apparent distinctions in the dimension of discretization [2] between the *information formats* utilized in different information presentation solutions are mainly classified into two categories: uses a spatialized floating panel to convey all the information about the product and presents different information directly and dispersedly in the corresponding position of the 3D product model. *Presentation mode* is associated with the ease with which product information can be interpreted to facilitate the shopping process, including a short description, long description, image, animation, and diagram. Furthermore, we determine the different interactions of each design case in the collection, in which rotation and scaling enable users to manipulate 3D products, and click enables users to trigger various information or product displays. According to Alves et al. [1], no matter how the 3D information is arranged, it has some relationship with the arrangement of the product. We further investigated the *spatial relation* of those design cases in the collection. It mainly includes three types: 1) attached and parallel to the product's surface; 2) suspended in the 3D environment but not parallel to any surface; 3) attached and parallel to other surfaces in space.

2.3 Analyzing and Exploring a Design Space

In the abovementioned work, we constructed an initial design space by deconstructing design cases to reflect our understanding of the collection's 3D product information presentation solutions. We continually analyzed and explored the current design space to narrow the design space to a manageable framework and determine whether the constituent elements in each dimension are sufficiently fine-grained and representative. After iterations, we condensed the design space into three dimensions: **visual presentation**, **user interaction**, and **spatial relationship**, as shown in Figure 1.

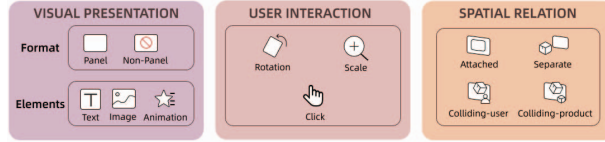


Figure 1: Design space

Since both information presentation mode and information format belong to the design features of information, we combine these two dimensions as two parts of the visual presentation of information. The information format consists of two forms as mentioned above. To simplify and classify the description of the design space, we classified the form of short and long descriptions in the initial design space as text. Moreover, the diagram element is ultimately removed. The visual presentation dimension of each 3D information presentation solution is composed of these two parts.

The interaction operations described in the initial design space are essential and fundamental in terms of user interaction, including the three basic interactions we mentioned above: rotate, zoom and click.

We sought to normalize the representation of the constituent elements in the dimension of spatial relation. According to Lee et al. [3], which explore the relationship between 3D visualization content and any particular surface in the environment, we respectively redefined the components of spatial relation dimension as follows: 'Attached,' which the information aligned parallel to the product surface; 'Colliding,' where the information is suspended in the 3D space, but its plane intersects with the product surface; 'Separate,' where the information is not related to the product surface in this relationship. Furthermore, we divide the 'colliding' element into the user and product centers according to two perspectives. The user center means that regardless of how the product is rotated, the information always faces the user. Conversely, the product center refers to the product as a reference point, and the information is always followed by the product.

2.4 Produce Solutions From Design Space

We created our design space of 3D product information to guide our research. To make our work more persuasive, we intend to generate a set of recommendations for designing 3D product information that improves user's shopping experience through user research. However, since the existing designs in the collection are not complete applications of the design space, we used the method of 'filling gaps' proposed by Ens [2] to produce new designs.

To sum up, nine feasible information presentation combinations were ultimately developed for user research, including solutions already existing in the collection and new combinations created through filling gaps. As shown in Figure 2, it is a list of the nine 3D product information combinations and their constituent parts. It is believed that these nine combinations are reasonable creations that can represent different combinations of design space.

3 RESEARCH

In this study, we aim to independently evaluate the individual effects of different 3D information presentation solutions on user's shopping experience. Specifically, we hope to answer the following questions through user study in follow-up work:

- **RQ1:** How efficient are different information presentation solutions in the task of product information retrieval? And how do they affect users' cognition and memory of product information?

	Space Relation	Visual Presentation	User Interaction
1	Attached	Non-Panel + Text Image Animation	Rotation Scale Click
2	Colliding-product	Panel + Text	Click
3	Colliding-product	Non-Panel + Text Image Animation	Rotation Scale Click
4	Colliding-product	Non-Panel + Text	Rotation Scale
5	Colliding-user	Panel + Text	Click
6	Colliding-user	Non-Panel + Text Image Animation	Rotation Scale Click
7	Colliding-user	Non-Panel + Text	Rotation Scale
8	Separate	Panel + Text	Click
9	Separate	Non-Panel + Text Image Animation	Click

Figure 2: The nine 3D product information combinations

- **RQ2:** How do different information presentation solutions affect the user's task load when browsing product information in a XR shopping application?
- **RQ3:** How do different information presentation solutions affect users' shopping experience and satisfaction with the interface of product information?

In order to better evaluate the user experience of information presentation solutions when users browse product information, we plan to design two tasks: the scenario test is used to examine the task performance for each solution, and the usability test evaluates the hedonic experience of the solutions.

4 CONCLUSION

The current solution of 3D product information in XR shopping applications may create a sense of fragmentation and an unsatisfactory user experience in terms of information presentation. There is no theoretical instruction in this area. Thus we deconstructed and refined a design space with three dimensions from a collection of existing designs and created new solutions. Successful completion of this research will provide some thoughts and recommendations for designing 3D product information presentations in the XR shopping application.

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